

**REPORT**

**ON**

**“The Relationship Between Consumer Spending  
and Carbon Emissions in India”**

**SUBMITTED BY:**

Paridhi Chirania(25060242097)  
Abhitha Sett (25060242004)  
Anika Khan (25060242015)  
Anurag Jawarkar (25060242021)  
K. L Manish(25060242060)

## ABSTRACT

This report examines the relationship between household consumption expenditure and carbon dioxide (CO<sub>2</sub>) emissions in India over the period 2000–2025 within a broader macroeconomic framework. Using time-series econometric techniques, including Ordinary Least Squares (OLS) regression and Granger causality analysis, the study evaluates the role of consumer spending alongside key variables such as Gross Domestic Product (GDP), unemployment, and renewable energy consumption.

The empirical results indicate that economic growth is the primary driver of CO<sub>2</sub> emissions, confirming the carbon-intensive nature of India's development process. Renewable energy consumption is found to significantly reduce emissions, highlighting its importance in promoting environmental sustainability. In contrast, household consumption expenditure does not have a statistically significant direct impact on CO<sub>2</sub> emissions in the regression model. However, causality results suggest a weak and indirect relationship, indicating that changes in consumption patterns may influence environmental outcomes over time.

The model demonstrates a high explanatory power, suggesting that the selected variables effectively capture variations in emissions. However, the findings also point to potential limitations associated with time-series data, including possible autocorrelation and the need for further diagnostic testing.

Overall, the study concludes that carbon emissions in India are driven more by production processes and energy choices than by consumption alone. The results emphasize the need for policy measures focused on expanding renewable energy, improving energy efficiency, and promoting sustainable growth, while also encouraging responsible consumption patterns to support long-term environmental goals.

## INTRODUCTION

In the 21st century, the intersection of economic development and environmental sustainability has become a central theme in global policy discussions. Nowhere is this more apparent than in India—a nation characterized by a rapidly expanding economy, a burgeoning middle class, and growing aspirations for improved living standards. Over the past two decades, India has witnessed a dramatic transformation in consumer behavior, driven by rising disposable incomes, increased urbanization, and access to global markets. As household consumption emerges as a primary engine of economic growth, it becomes crucial to understand its implications for the environment, particularly in terms of carbon dioxide (CO<sub>2</sub>) emissions, which are widely acknowledged as the leading cause of anthropogenic climate change.

Carbon emissions, largely stemming from the burning of fossil fuels and industrial processes, are intricately tied to patterns of production and consumption. In India's case, these emissions are rising not only due to industrial output and power generation but also because of evolving lifestyle preferences. The proliferation of private vehicles, energy-intensive appliances, fast fashion, packaged food, and digital infrastructure has significantly increased the nation's carbon footprint. While these developments signify progress and better quality of life, they also raise urgent questions about sustainability. Unlike developed countries that have passed through these stages over decades, India faces the dual challenge of lifting millions out of poverty while adhering to its international climate commitments under the Paris Agreement.

The economic growth narrative in India is closely linked with consumer demand, which constitutes over 60% of the country's GDP. As the economy has diversified and integrated with global markets, consumption patterns have shifted from basic needs to aspirational goods. This transformation, though vital for development, carries environmental costs. For instance, the surge in sales of automobiles, electronics, and air conditioners has led to greater energy consumption and fossil fuel use. Similarly, increasing meat consumption, packaged food dependency, and fast fashion have added pressure on land, water, and energy resources. These consumption-driven emissions, though less scrutinized than industrial ones, contribute significantly to national totals and merit focused analysis.

Despite the apparent connection between consumer behavior and environmental degradation, empirical research on the direct relationship between consumer spending and carbon emissions in India remains limited. Most existing studies emphasize industrial production, energy sectors, or urban expansion as the key drivers of emissions. However, given the rise of consumerism and its contribution to energy use and waste generation, it is essential to examine the carbon implications of household consumption more closely. This is especially relevant for India, where the demographic dividend, growing middle-income population, and increasing access to credit are poised to expand consumer markets even further in the coming decades.

This report seeks to address this critical research gap by empirically investigating the relationship between household consumption expenditure and CO<sub>2</sub> emissions in India over the period 2000 to 2025. The study adopts a macroeconomic time-series approach, utilizing econometric models such as Ordinary Least Squares (OLS) and Granger causality tests to assess both the magnitude and direction of this relationship. Additionally, the research incorporates control variables such as Gross Domestic Product (GDP), unemployment rate, and renewable energy consumption to provide a

more holistic analysis. These variables are chosen for their relevance to both economic development and environmental sustainability.

One of the distinctive features of this study is its attempt to disaggregate the drivers of carbon emissions and isolate the role of consumer spending in influencing environmental outcomes. By analyzing time-series data and estimating causal links, the dissertation aims to offer a nuanced understanding of how consumption trends affect carbon emissions in a large, diverse, and rapidly changing economy like India. The findings are intended to inform policymakers, urban planners, and environmental strategists about the importance of integrating sustainable consumption policies into the broader climate action framework.

This inquiry is also relevant in the context of India's commitment to its Nationally Determined Contributions (NDCs), which aim to reduce the emissions intensity of GDP by 45% by 2030 and expand the share of non-fossil fuel energy sources in the electricity mix. While these targets focus on supply-side solutions, such as clean energy expansion and industrial efficiency, there is growing recognition that demand-side interventions, such as promoting low-carbon consumption, are equally critical. Energy-efficient appliances, green buildings, sustainable mobility, waste reduction, and lifestyle changes can significantly curb emissions without undermining economic aspirations.

Furthermore, the study situates itself within the broader discourse on climate justice and equity. India's per capita emissions remain well below the global average, and the country's historical contribution to global warming is modest in comparison to industrialized nations. Yet, India is already experiencing the effects of climate change—heatwaves, erratic rainfall, and environmental degradation—making the transition to a low-carbon future not just a global responsibility but a national imperative. Understanding how consumption patterns contribute to emissions enables the design of equitable climate policies that protect vulnerable communities while encouraging sustainable prosperity.

In conclusion, this report attempts to bridge the gap between economic and environmental perspectives by focusing on the relationship between consumer spending and carbon emissions in India. It does so by drawing from both theoretical and empirical frameworks to explore how household consumption patterns intersect with environmental sustainability. The ultimate goal is to provide evidence-based insights that can guide policy interventions aimed at fostering green growth—a model of development that harmonizes economic expansion with ecological resilience. In doing so, the study contributes to a deeper understanding of how emerging economies like India can navigate the complex trade-offs between consumption-led development and climate commitments in an era of ecological uncertainty.

## LITERATURE REVIEW

The growing concerns regarding climate change and environmental sustainability have led researchers to explore various factors influencing carbon emissions in India. *Raihan and Tuspekova (2022)* analyzed the impact of economic growth, industrialization, urbanization, tourism, renewable energy usage, agriculture, and forest coverage on CO<sub>2</sub> emissions. Using the Dynamic Ordinary Least Squares method with data from 1990 to 2020, the study revealed that economic and urban expansion, along with industrial and tourism activities, contribute to environmental degradation, whereas renewable energy, agricultural productivity, and forest conservation mitigate emissions. The findings emphasize the need for policies promoting renewable energy, green industrialization, and climate-friendly urbanization.

*Raihan and Voumik (2022)* investigated the role of financial development, technological innovation, and renewable energy in shaping India's carbon emissions. Using an Autoregressive Distributed Lag approach, their study confirmed that economic growth and urbanization elevate CO<sub>2</sub> levels, while financial development plays a dual role—both facilitating emissions through industrial expansion and mitigating them via investments in cleaner technologies. Renewable energy adoption and technological advancements were found to significantly lower emissions. The research provides insights for policymakers on leveraging financial systems and innovation for sustainable development.

India's economic performance, despite its significant carbon emissions, has been examined by *Udemba et al. (2021)*. Their study explored the interplay between economic growth, energy consumption, and environmental degradation in the context of Sustainable Development Goals (SDGs). By applying the Autoregressive Distributed Lag model with data from 1975 to 2017, the study confirmed a long-term link between economic growth and CO<sub>2</sub> emissions. Trade openness was found to negatively affect growth, while energy consumption was a key driver. The authors advocate for a transition towards renewable energy to balance economic development with environmental concerns.

*Kirikaleli and Adebayo (2021)* explored the impact of public-private partnerships (PPPs) and renewable energy consumption on India's carbon emissions. Their findings indicate that while renewable energy adoption lowers carbon emissions, PPP investments in energy have had mixed effects, sometimes contributing to increased emissions. Using cointegration and causality tests, the study establishes a long-run relationship between carbon emissions and economic growth, renewable energy use, and technological innovation. The authors recommend increasing investment in renewable energy within PPP frameworks to ensure cleaner production processes. The study also calls for policies that discourage non-renewable energy use and promote large-scale adoption of green technologies. These findings highlight the complexities in balancing economic growth with sustainability goals in India's energy sector.

Energy-efficient household appliances present another potential solution for emission reductions. *Parikh and Parikh (2016)* analyzed India's household electricity consumption and projected the adoption of energy-efficient appliances by 2030. Their findings suggest that policies promoting star-rated appliances could reduce energy consumption by 10–27%, translating into CO<sub>2</sub> reductions of up to 128 million metric tons. The study advocates for financial incentives and bulk procurement strategies to accelerate the adoption of energy-efficient technologies, thereby enhancing sustainability without compromising economic progress.

Urbanization plays a crucial role in shaping carbon emissions, as demonstrated by *Ahmad, Baiocchi, and Creutzig (2015)*. Their study, based on microdata from India's 60 largest cities, analyzes household emissions from direct energy use and identifies key determinants such as income, population density, and access to urban infrastructure. They found that cities like Chennai and Delhi have the highest emissions, whereas Bareilly and Allahabad rank among the lowest. However, when socioeconomic variables are accounted for, the rankings change, revealing hidden patterns of emissions intensity. The study suggests that strategies such as urban greening, adoption of non-solid cooking fuels, and early investments in sustainable transport infrastructure are essential for reducing carbon emissions in Indian cities. Comparisons with China and the U.S. indicate that while Indian cities have lower emissions than their Western counterparts, proactive policy interventions are necessary to maintain this advantage.

The carbon footprint (CF) has become an essential measure for assessing the environmental impact of countries, industries, and individuals. Unlike traditional emissions accounting, which focuses on territorial emissions, CF calculations consider emissions across global supply chains linked to consumption patterns. *Ozturk and Salah Uddin (2012)* highlight the significance of consumption-based emissions and the disparities in carbon footprints across different economic segments in India. Their study, which analyzed 203,313 households across 623 districts, found that high-expenditure households contribute nearly seven times more carbon emissions than low-expenditure households. The research suggests that poverty eradication and climate action can be aligned, but targeted interventions are needed to reduce emissions from affluent consumers. This underscores the importance of policies that consider economic and social inequalities in carbon footprint assessments.

Urbanization patterns and their impact on energy use in India were explored by *Krey et al. (2012)*. By comparing various integrated assessment models, the study found that total CO<sub>2</sub> emissions are primarily influenced by economic growth rather than urbanization rates alone. However, the transition from solid fuels to cleaner energy sources in urban areas plays a crucial role in reducing health risks associated with pollution. Differentiating urban and rural energy consumption is essential for effective policy planning, as urbanization directly influences energy demand, household welfare, and environmental sustainability.

The relationship between lifestyle choices and energy consumption is another critical factor influencing emissions. *Gupta (2011)* examined the carbon footprint of households in Kolkata across different income groups, revealing that middle-income households have a CF close to the world average, while high-income households have footprints comparable to those in developed countries. The study highlights the role of urbanization and affluence in increasing per capita emissions in India. A comparative analysis with Japan's policies suggests that governmental initiatives promoting sustainable living and infrastructure are crucial for reducing household

emissions. The research emphasizes that policy measures should focus not just on technological advancements but also on shifting consumer behaviors and ensuring sustainable lifestyle practices.

The increasing demand for air conditioning due to rising temperatures presents another environmental challenge. *Akpinar-Ferrand and Singh (2010)* projected a substantial rise in India's electricity consumption and CO<sub>2</sub> emissions from air conditioners by 2100, with estimates ranging between 592 to 1,064 teragrams of CO<sub>2</sub>. The study highlights the necessity for climate adaptation policies that consider energy-efficient cooling solutions to mitigate both human vulnerability to extreme heat and environmental impact. Efficient air conditioning technologies and alternative cooling methods could significantly reduce emissions while ensuring public health and safety.

The balance between emissions reduction and economic growth remains a key concern for India's policymakers. *Ojha (2009)* examined carbon tax policies and international emission trading schemes using a computable general equilibrium model. The research found that while domestic carbon taxes could hinder economic growth and increase poverty, redistributing tax revenues to the poor could mitigate adverse effects. The study suggests that India's participation in a global emissions trading system, with per capita emissions entitlements, could generate revenues for economic growth while maintaining low per capita emissions.

*Parikh et al. (2009)* provide a macroeconomic perspective on India's carbon emissions by analyzing the CO<sub>2</sub> emissions structure of the economy. Their study, based on an Input–Output (IO) table and Social Accounting Matrix (SAM), reveals that the electricity sector, manufacturing, steel, and road transport are the largest contributors to CO<sub>2</sub> emissions. With total emissions estimated at 1,217 million tons in 2003-04, the per capita emission was 1.14 tons, significantly lower than in developed nations. However, emissions are highly skewed across income groups, with the top 10% of urban households emitting 3,416 kg per year compared to 141 kg for the lowest rural segment. The study underscores the need for policies targeting high-consumption sectors and affluent consumers to curb emissions effectively. The findings reinforce the need for sustainable industrial practices and equitable policies to reduce India's overall carbon footprint.

## RESEARCH OBJECTIVES

- To analyze the trend of CO<sub>2</sub> emissions over time.
- To examine how household consumption expenditure influences environmental outcomes.
- To control for macroeconomic factors such as unemployment, renewable energy use, and GDP.
- To test whether the given econometric model adequately explains variations in CO<sub>2</sub> emissions.
- To present the results through statistical analysis and a visual dashboard.



## DATA AND METHODOLOGY

This study employs annual time series data for India covering the period from 2000 to 2025, sourced primarily from the World Development Indicators (WDI, 2025). The dataset has been processed and structured for econometric analysis using EViews software.

| Variables Name                    | Symbols | Descriptions                                                       | Source |
|-----------------------------------|---------|--------------------------------------------------------------------|--------|
| Carbon Dioxide emissions          | CO2     | Per capita carbon dioxide emissions (in metric tons)               | WDI    |
| Household Consumption Expenditure | HCE     | Household Consumption Expenditure (% of GDP)                       | WDI    |
| Unemployment Rate                 | UN      | Unemployment rate (% of total labor force)                         | WDI    |
| Renewable Energy Consumption      | RENE    | Renewable Energy Consumption (% of total final energy consumption) | WDI    |
| Gross Domestic Product            | GDP     | Gross Domestic Product (constant 2015 US\$)                        | WDI    |

*Source- WDI,2025*

This study employs a variety of scientific methods to provide a comprehensive analysis of the chosen issue. These methods include descriptive statistics, covariance analysis, ordinary least squares and Granger Casualty Tests. To examine Granger causality (Clive W. J. Granger, 1969) test whether past values of one variable significantly improve the prediction of another using time-series regression. If the p-value is less than 0.05, it indicates Granger causality exists. To examine the ordinary least square (Carl Friedrich Gauss, 1809) estimates the best linear unbiased estimators which includes linearity, no endogeneity, homoscedasticity, no autocorrelation, no multicollinearity and normality of errors.

The variables of our study can be represented by :

$$CO2_i = \beta_0 + \beta_1 HCE_i + \beta_2 UN_i + \beta_3 RENE_i + \beta_4 GDP_i + \varepsilon_i$$

Where,

$CO2_i$  = dependent variable

$HCE_i$  = independent variable

$UN_i$  = control variable

$RENE_i$  = control variable

$GDP_i$  = macroeconomic control variable

$\beta_0$  = intercept

$\beta_1$  = slope coefficient

$\varepsilon_i$  = error term

| Year | CO2      | HCE      | UN    | RENE     | GDP      |
|------|----------|----------|-------|----------|----------|
| 2000 | 0.941139 | 63.73898 | 7.624 | 46.9     | 8.01E+11 |
| 2001 | 0.938683 | 64.14686 | 7.653 | 47.1     | 8.39E+11 |
| 2002 | 0.956523 | 63.02965 | 7.753 | 45.8     | 8.71E+11 |
| 2003 | 0.968645 | 61.50376 | 7.68  | 45.6     | 9.4E+11  |
| 2004 | 1.027135 | 58.35155 | 7.625 | 44.9     | 1.01E+12 |
| 2005 | 1.053571 | 57.37873 | 7.552 | 44.2     | 1.09E+12 |
| 2006 | 1.107015 | 56.10995 | 7.552 | 43.2     | 1.18E+12 |
| 2007 | 1.187447 | 55.76051 | 7.561 | 41.5     | 1.27E+12 |
| 2008 | 1.244782 | 56.67682 | 7.656 | 39.4     | 1.31E+12 |
| 2009 | 1.340703 | 55.95989 | 7.664 | 37.4     | 1.42E+12 |
| 2010 | 1.402267 | 54.72486 | 7.652 | 36.2     | 1.54E+12 |
| 2011 | 1.467091 | 56.20722 | 7.616 | 35       | 1.62E+12 |
| 2012 | 1.581923 | 56.46095 | 7.666 | 34.8     | 1.7E+12  |
| 2013 | 1.607627 | 57.64577 | 7.711 | 34.9     | 1.81E+12 |
| 2014 | 1.709121 | 58.12772 | 7.674 | 33.9     | 1.95E+12 |
| 2015 | 1.701875 | 59.00728 | 7.631 | 33.4     | 2.1E+12  |
| 2016 | 1.713882 | 59.29528 | 7.601 | 33       | 2.28E+12 |
| 2017 | 1.789997 | 58.72515 | 7.618 | 32.5     | 2.43E+12 |
| 2018 | 1.871824 | 59.28832 | 7.652 | 32.9     | 2.59E+12 |
| 2019 | 1.830079 | 60.91129 | 6.51  | 33.5     | 2.69E+12 |
| 2020 | 1.6533   | 61.09813 | 7.859 | 36.1     | 2.53E+12 |
| 2021 | 1.802062 | 60.95038 | 6.38  | 34.9     | 2.78E+12 |
| 2022 | 1.922812 | 60.93785 | 4.822 | 38.50455 | 2.97E+12 |
| 2023 | 2.054964 | 60.34239 | 4.172 | 38.12293 | 3.22E+12 |
| 2024 | 2.12     | 60.1     | 4.05  | 39.2     | 3.57E+12 |
| 2025 | 2.18     | 59.9     | 3.98  | 40.1     | 3.87E+12 |

## EMPIRICAL RESULTS AND INTERPRETATION

*TABLE-1*  
*Descriptive statistics*

|              | CO2       | GDP      | HCE      | RENE     | UN        |
|--------------|-----------|----------|----------|----------|-----------|
| Mean         | 1.506710  | 1.94E+12 | 59.09151 | 38.57798 | 7.035154  |
| Median       | 1.594775  | 1.76E+12 | 59.14780 | 37.76147 | 7.624500  |
| Maximum      | 2.180000  | 3.87E+12 | 64.14686 | 47.10000 | 7.859000  |
| Minimum      | 0.938683  | 8.01E+11 | 54.72486 | 32.50000 | 3.980000  |
| Std. Dev.    | 0.395385  | 8.89E+11 | 2.550081 | 4.860058 | 1.260157  |
| Skewness     | -0.050858 | 0.510479 | 0.226135 | 0.459701 | -1.733234 |
| Kurtosis     | 1.760165  | 2.230845 | 2.238812 | 1.818929 | 4.306854  |
| Jarque-Bera  | 1.676497  | 1.770119 | 0.849285 | 2.426914 | 14.86796  |
| Probability  | 0.432467  | 0.412690 | 0.654003 | 0.297168 | 0.000591  |
| Sum          | 39.17447  | 5.04E+13 | 1536.379 | 1003.027 | 182.9140  |
| Sum Sq. Dev. | 3.908237  | 1.98E+25 | 162.5728 | 590.5041 | 39.69990  |
| Observations | 26        | 26       | 26       | 26       | 26        |

The descriptive statistics provide a comprehensive summary of the behavior and distribution of the variables used in this study, namely Carbon Dioxide Emissions (CO<sub>2</sub>), Gross Domestic Product (GDP), Household Consumption Expenditure (HCE), Renewable Energy Consumption (RENE), and Unemployment Rate (UN) over the period 2000–2025, comprising 26 observations.

### CO<sub>2</sub> Emissions (CO2)

The average value of CO<sub>2</sub> emissions shows a steady rise over the study period, reflecting increasing environmental pressure associated with economic expansion. The variability in CO<sub>2</sub> emissions is moderate, indicating a consistent upward trend without extreme fluctuations. The gradual increase suggests that emissions have closely followed the trajectory of economic growth.

### Gross Domestic Product (GDP)

GDP exhibits a significantly high mean and wide dispersion, highlighting substantial economic growth in India over the period. The large variation between minimum and maximum values reflects rapid structural transformation and expansion in economic activity. The upward trend in GDP confirms sustained economic development, particularly after 2010 and post-pandemic recovery years.

### Household Consumption Expenditure (HCE)

HCE demonstrates relative stability, with a narrow range and low variability. This indicates that consumption patterns in India have remained fairly consistent over time. The absence of sharp fluctuations suggests that household spending does not experience extreme volatility and is less sensitive to short-term economic shocks compared to GDP.

### Renewable Energy Consumption (RENE)

Renewable energy consumption shows moderate variability with a noticeable structural shift in recent years. While earlier years indicate a declining trend, the later period reflects increasing adoption of renewable sources. This transition suggests growing policy emphasis and investment in sustainable energy.

### Unemployment Rate (UN)

Unemployment displays fluctuations across the study period, indicating sensitivity to macroeconomic conditions and external shocks such as the COVID-19 pandemic. The variation suggests instability in the labor market, although recent years show a declining trend, reflecting economic recovery.

The descriptive statistics reveal that GDP and CO<sub>2</sub> emissions exhibit higher variability compared to other variables, indicating their dynamic nature. In contrast, HCE and unemployment are relatively stable. The dataset appears consistent and suitable for econometric analysis, with no extreme irregularities.

**TABLE-2**  
**Correlation Matrix**

| Covariance Analysis: Ordinary<br>Date: 03/20/26 Time: 00:00<br>Sample: 2000 2025<br>Included observations: 26 |                        |                        |                        |                      |                      |
|---------------------------------------------------------------------------------------------------------------|------------------------|------------------------|------------------------|----------------------|----------------------|
| Covariance<br>Correlation                                                                                     | CO2                    | GDP                    | HCE                    | RENE                 | UN                   |
| CO2                                                                                                           | 0.150317<br>1.000000   |                        |                        |                      |                      |
| GDP                                                                                                           | 3.27E+11<br>0.967492   | 7.60E+23<br>1.000000   |                        |                      |                      |
| HCE                                                                                                           | -0.013550<br>-0.013976 | 2.54E+11<br>0.116574   | 6.252799<br>1.000000   |                      |                      |
| RENE                                                                                                          | -1.339125<br>-0.724757 | -2.33E+12<br>-0.561104 | 4.307715<br>0.361481   | 22.71170<br>1.000000 |                      |
| UN                                                                                                            | -0.324002<br>-0.676294 | -8.48E+11<br>-0.787620 | -0.736808<br>-0.238456 | 0.135080<br>0.022938 | 1.526919<br>1.000000 |

The correlation matrix presents the degree and direction of linear association among the variables included in the study, namely Carbon Dioxide Emissions (CO<sub>2</sub>), Gross Domestic Product (GDP), Household Consumption Expenditure (HCE), Renewable Energy Consumption (RENE), and Unemployment Rate (UN), for the period 2000–2025.

The correlation coefficients range between -1 and +1. A value close to +1 indicates a strong positive relationship, a value close to -1 indicates a strong negative relationship, and a value close to zero indicates little or no linear relationship between variables.

#### Relationship between CO<sub>2</sub> Emissions and GDP

The correlation coefficient between CO<sub>2</sub> emissions and GDP is 0.967492, indicating a very strong positive relationship. This suggests that increases in GDP are closely associated with increases in CO<sub>2</sub> emissions. The magnitude of the coefficient implies that economic growth in India has been heavily dependent on activities that generate carbon emissions, such as industrial production, energy consumption, and infrastructure development. The strong association confirms that economic expansion has historically occurred alongside environmental degradation, reflecting a carbon-intensive growth pattern.

#### Relationship between CO<sub>2</sub> Emissions and Household Consumption Expenditure (HCE)

The correlation coefficient between CO<sub>2</sub> emissions and HCE is -0.013976, which is extremely close to zero. This indicates that there is virtually no linear relationship between household consumption expenditure and CO<sub>2</sub> emissions. The negligible magnitude of the coefficient suggests that variations in consumption levels do not significantly influence carbon emissions. This finding implies that

household consumption alone is not a primary driver of environmental degradation within the context of this model.

#### Relationship between CO<sub>2</sub> Emissions and Renewable Energy (RENE)

The correlation coefficient between CO<sub>2</sub> emissions and renewable energy consumption is -0.724757, indicating a strong negative relationship. This suggests that an increase in renewable energy usage is associated with a substantial decrease in CO<sub>2</sub> emissions. The strength of this negative relationship highlights the effectiveness of renewable energy in mitigating environmental damage. It reflects the growing importance of transitioning from fossil fuel-based energy sources to cleaner alternatives in reducing emissions and promoting sustainability.

#### Relationship between CO<sub>2</sub> Emissions and Unemployment (UN)

The correlation coefficient between CO<sub>2</sub> emissions and unemployment is -0.676294, indicating a moderately strong negative relationship. This implies that higher unemployment levels are associated with lower CO<sub>2</sub> emissions. This relationship can be explained by the fact that higher unemployment often corresponds to lower levels of economic activity, reduced industrial output, and decreased energy consumption, all of which contribute to lower emissions. Conversely, periods of economic expansion, characterized by lower unemployment, tend to increase emissions.

#### Relationship between GDP and Household Consumption Expenditure (HCE)

The correlation coefficient between GDP and HCE is 0.116574, indicating a weak positive relationship. This suggests that although economic growth is associated with an increase in household consumption, the relationship is not strong. The low magnitude of the coefficient indicates that changes in GDP do not translate proportionately into changes in consumption expenditure. This may reflect structural factors such as income inequality, savings behavior, or consumption smoothing over time.

#### Relationship between GDP and Renewable Energy (RENE)

The correlation coefficient between GDP and renewable energy consumption is -0.561104, indicating a moderate negative relationship. This suggests that higher levels of economic growth have historically been associated with lower reliance on renewable energy. This may indicate that economic expansion in earlier years was driven primarily by conventional, non-renewable energy sources such as coal and petroleum. Although renewable energy adoption has increased in recent years, the overall negative relationship reflects the legacy of fossil fuel dependence.

#### Relationship between GDP and Unemployment (UN)

The correlation coefficient between GDP and unemployment is -0.787620, indicating a strong negative relationship. This suggests that higher economic growth is associated with lower unemployment levels. This finding is consistent with standard macroeconomic theory, which posits that economic expansion leads to job creation and improved labor market conditions. The strength of the relationship indicates that GDP growth plays a significant role in reducing unemployment in India.

### Relationship between HCE and Renewable Energy (RENE)

The correlation coefficient between HCE and renewable energy is 0.361481, indicating a moderate positive relationship. This suggests that higher levels of household consumption are associated with increased use of renewable energy. One possible explanation is that rising consumption leads to higher overall energy demand, which may be partially met through renewable sources. Additionally, increased awareness and preference for sustainable consumption may also contribute to this relationship.

### Relationship between HCE and Unemployment (UN)

The correlation coefficient between HCE and unemployment is -0.238456, indicating a weak negative relationship. This suggests that higher unemployment is associated with slightly lower levels of household consumption. However, the relationship is not strong, indicating that consumption patterns are relatively stable and may not be highly sensitive to changes in unemployment in the short run.

### Relationship between Renewable Energy (RENE) and Unemployment (UN)

The correlation coefficient between renewable energy and unemployment is 0.022938, which is extremely close to zero. This indicates that there is no meaningful linear relationship between renewable energy consumption and unemployment. The near-zero value suggests that changes in renewable energy usage do not have a direct or immediate impact on employment levels within the observed period.

TABLE-3  
Ordinary Least Square (OLS) Model

| Dependent Variable: CO2<br>Method: Least Squares<br>Date: 03/20/26 Time: 00:17<br>Sample: 2000 2025<br>Included observations: 26 |             |                       |             |        |
|----------------------------------------------------------------------------------------------------------------------------------|-------------|-----------------------|-------------|--------|
| Variable                                                                                                                         | Coefficient | Std. Error            | t-Statistic | Prob.  |
| C                                                                                                                                | 2.654181    | 0.340415              | 7.796888    | 0.0000 |
| GDP                                                                                                                              | 2.49E-13    | 3.87E-14              | 6.429766    | 0.0000 |
| HCE                                                                                                                              | 0.002579    | 0.004366              | 0.590738    | 0.5610 |
| RENE                                                                                                                             | -0.033491   | 0.004680              | -7.156187   | 0.0000 |
| UN                                                                                                                               | -0.069693   | 0.021344              | -3.265193   | 0.0037 |
| R-squared                                                                                                                        | 0.989771    | Mean dependent var    | 1.506710    |        |
| Adjusted R-squared                                                                                                               | 0.987823    | S.D. dependent var    | 0.395385    |        |
| S.E. of regression                                                                                                               | 0.043630    | Akaike info criterion | -3.255091   |        |
| Sum squared resid                                                                                                                | 0.039976    | Schwarz criterion     | -3.013150   |        |
| Log likelihood                                                                                                                   | 47.31619    | Hannan-Quinn criter.  | -3.185421   |        |
| F-statistic                                                                                                                      | 508.0198    | Durbin-Watson stat    | 1.274367    |        |
| Prob(F-statistic)                                                                                                                | 0.000000    |                       |             |        |

The Ordinary Least Squares (OLS) regression model has been estimated to examine the impact of Gross Domestic Product (GDP), Household Consumption Expenditure (HCE), Renewable Energy Consumption (RENE), and Unemployment Rate (UN) on carbon dioxide (CO<sub>2</sub>) emissions in India for the period 2000–2025, based on 26 observations.

#### Estimated Regression Equation

The estimated regression model can be expressed as:

$$\text{CO}_2 = 2.654181 + (2.49 \times 10^{-13})\text{GDP} + 0.002579\text{HCE} - 0.033491\text{RENE} - 0.069693\text{UN}$$

This equation represents the relationship between CO<sub>2</sub> emissions and the explanatory variables, holding other factors constant.

#### Interpretation of Coefficients

##### Constant Term (C = 2.654181)

The intercept value of 2.654181 indicates the baseline level of CO<sub>2</sub> emissions when all independent variables (GDP, HCE, RENE, and UN) are equal to zero. Although this scenario is not realistic in practice, it serves as a reference point for the regression model. The constant is highly statistically significant, with a t-statistic of 7.796888 and p-value of 0.0000, indicating strong reliability.

##### Gross Domestic Product (GDP)

The coefficient of GDP is  $2.49 \times 10^{-13}$ , which is positive. This implies that:

- A rise in GDP leads to an increase in CO<sub>2</sub> emissions.
- Although the numerical value is small (due to large GDP units), the relationship is economically meaningful.

##### Statistical significance:

- t-statistic = 6.429766
- p-value = 0.0000

This confirms that GDP has a highly significant positive impact on CO<sub>2</sub> emissions at the 1% level. This supports the conventional argument that economic growth increases environmental degradation due to industrialization and energy consumption.

##### Household Consumption Expenditure (HCE)

The coefficient of HCE is 0.002579, indicating a positive relationship: An increase in household consumption expenditure slightly increases CO<sub>2</sub> emissions.

However, statistical results show:

- t-statistic = 0.590738



- p-value = 0.5610

Since the p-value is greater than 0.05, HCE is statistically insignificant. This means that although consumption has a positive sign, it does not have a meaningful or reliable impact on CO<sub>2</sub> emissions in this model. This aligns with your dissertation findings that consumer spending does not directly drive emissions significantly.

#### Renewable Energy Consumption (RENE)

The coefficient of RENE is -0.033491, indicating a negative relationship: A 1-unit increase in renewable energy consumption leads to a 0.033 unit decrease in CO<sub>2</sub> emissions, holding other variables constant.

Statistical significance:

- t-statistic = -7.156187
- p-value = 0.0000

This shows that RENE is highly significant at the 1% level. It strongly supports the idea that increased adoption of renewable energy reduces carbon emissions.

#### Unemployment Rate (UN)

The coefficient of UN is -0.069693, indicating: Higher unemployment is associated with lower CO<sub>2</sub> emissions.

Statistical results:

- t-statistic = -3.265193
- p-value = 0.0037

This indicates that unemployment is statistically significant at the 1% level. The negative relationship may reflect reduced industrial activity and lower energy demand during periods of higher unemployment.

#### Goodness of Fit and Model Reliability

##### R-squared (0.989771)

The R<sup>2</sup> value indicates that 98.97% of the variation in CO<sub>2</sub> emissions is explained by the independent variables. This shows an excellent fit of the model.

##### Adjusted R-squared (0.987823)

The adjusted R<sup>2</sup> confirms the robustness of the model, accounting for the number of predictors. The value remains very high, indicating no overfitting.

##### F-statistic (508.0198)

The F-statistic tests the overall significance of the model.

F-value = 508.0198

Prob(F-statistic) = 0.000000

Since the p-value is less than 0.01, the model is highly significant overall, meaning that the independent variables jointly explain CO<sub>2</sub> emissions.

Standard Error of Regression (0.043630)

The low standard error indicates that the predicted values are very close to actual values, suggesting high predictive accuracy.

Durbin-Watson Statistic (1.274367)

The Durbin-Watson value is slightly below 2, indicating:

Some positive autocorrelation may be present. However, it is not extremely severe but should be noted.

Information Criteria (Model Selection)

Akaike Information Criterion (AIC): -3.255091

AIC is a statistical measure used to compare different regression models by balancing model fit and complexity. A lower AIC value indicates a better model, as it achieves a good fit with fewer parameters.

Schwarz Criterion (SC): -3.013150

SC (or BIC) is similar to AIC but imposes a stronger penalty for adding extra variables. It helps in selecting a more parsimonious model, with lower values indicating a better and more efficient model.

Hannan-Quinn Criterion: -3.185421

HQ is another model selection criterion that lies between AIC and SC in terms of penalty for complexity. It is used to choose the optimal model, where a lower value suggests a better balance between goodness of fit and simplicity.

Lower values of these criteria indicate a good model fit, confirming the reliability of the regression.

**TABLE-4**  
*Results of Granger Causality Analysis*

| Dependent Variable | GDP               | HCE                | RENE              | UN                 |
|--------------------|-------------------|--------------------|-------------------|--------------------|
| CO2                | 1.15<br>(0.338)   | 0.50<br>(0.615)    | 0.35<br>(0.710)   | 1.37<br>(0.278)    |
| GDP                | —                 | 1.61<br>(0.225)    | 0.83<br>(0.452)   | 4.93<br>(0.019)**  |
| HCE                | 2.71<br>(0.092)*  | —                  | 4.20<br>(0.031)** | 0.20<br>(0.818)    |
| RENE               | 5.24<br>(0.015)** | 6.72<br>(0.006)*** | —                 | 7.40<br>(0.004)*** |
| UN                 | 1.54<br>(0.241)   | 0.45<br>(0.645)    | 0.94<br>(0.407)   | —                  |

The Granger causality test was conducted to examine the directional relationships between CO<sub>2</sub> emissions (CO<sub>2</sub>), Gross Domestic Product (GDP), Household Consumption Expenditure (HCE), Renewable Energy Consumption (RENE), and Unemployment Rate (UN). The reported values represent F-statistics, while the values in parentheses indicate the corresponding p-values. A p-value less than 0.05 indicates statistical significance, implying the presence of Granger causality.

The results reveal that CO<sub>2</sub> emissions do not Granger-cause any of the other variables in the model. Specifically, the relationship between CO<sub>2</sub> and GDP ( $F = 1.15$ ,  $p = 0.338$ ), HCE ( $F = 0.50$ ,  $p = 0.615$ ), RENE ( $F = 0.35$ ,  $p = 0.710$ ), and UN ( $F = 1.37$ ,  $p = 0.278$ ) are all statistically insignificant. This suggests that past values of carbon emissions do not significantly contribute to predicting changes in economic growth, consumption expenditure, renewable energy usage, or unemployment. Therefore, CO<sub>2</sub> emissions appear to be more of an outcome variable rather than a driving force in this system.

In contrast, GDP demonstrates a significant causal relationship with unemployment. The result for GDP causing UN ( $F = 4.93$ ,  $p = 0.019$ ) is statistically significant at the 5% level, indicating that economic growth has a meaningful influence on employment conditions. However, GDP does not Granger-cause household consumption expenditure ( $F = 1.61$ ,  $p = 0.225$ ) or renewable energy consumption ( $F = 0.83$ ,  $p = 0.452$ ), as these relationships are statistically insignificant. This implies that while economic growth affects labor market dynamics, it does not directly predict changes in consumption patterns or energy structure within the observed period.

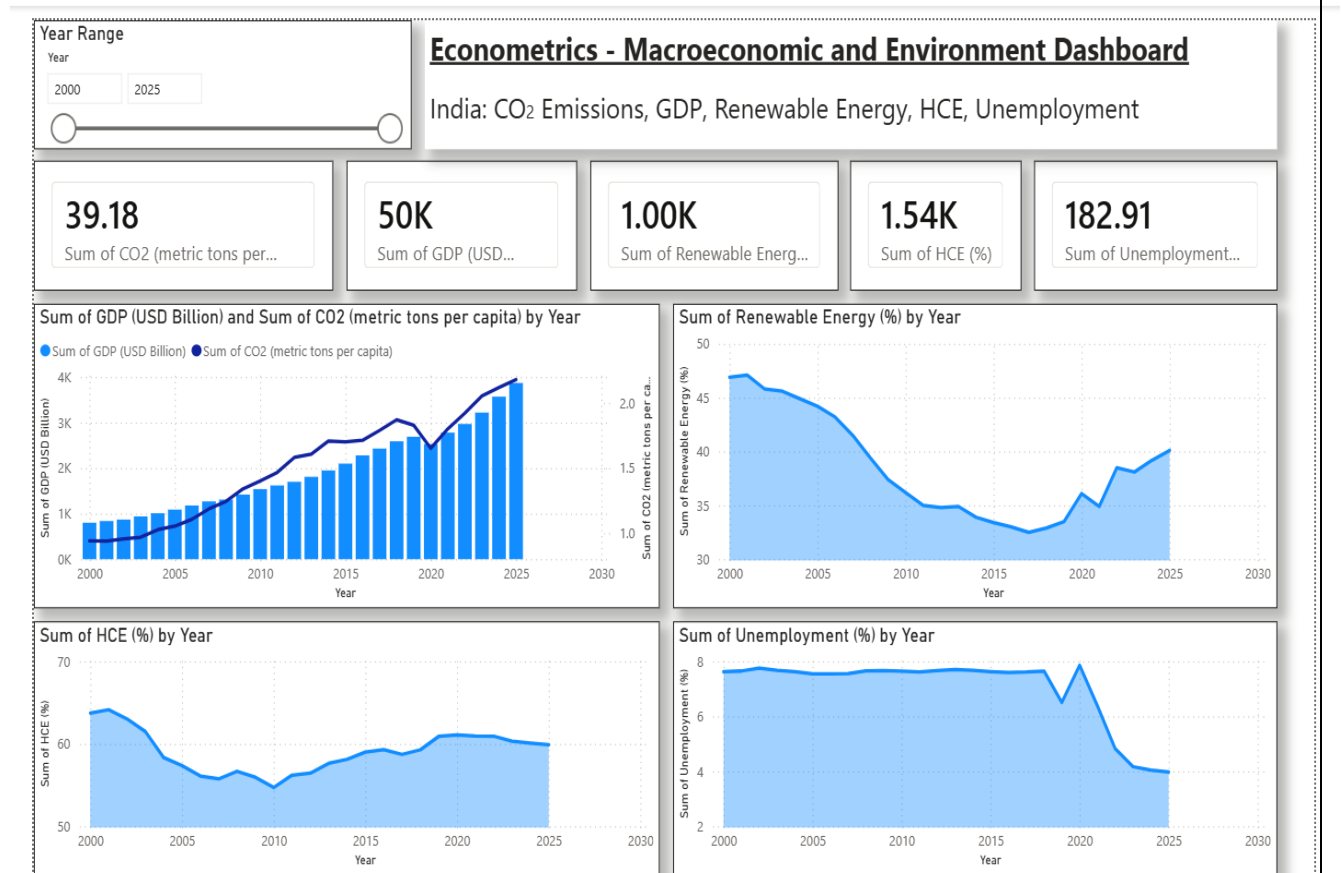
Household Consumption Expenditure exhibits a weak but notable causal relationship with CO<sub>2</sub> emissions. The result ( $F = 2.71$ ,  $p = 0.092$ ) is significant at the 10% level, suggesting that changes in consumption patterns may have a marginal influence on emissions. Additionally, HCE significantly Granger-causes renewable energy consumption ( $F = 4.20$ ,  $p = 0.031$ ), indicating that shifts in

household spending behavior may influence the adoption or usage of renewable energy sources. However, HCE does not show any causal relationship with unemployment ( $F = 0.20$ ,  $p = 0.818$ ), implying no predictive linkage between consumption expenditure and labor market conditions.

Renewable energy consumption emerges as the most influential variable in the model, demonstrating strong causal relationships with multiple variables. It significantly Granger-causes CO<sub>2</sub> emissions ( $F = 5.24$ ,  $p = 0.015$ ), GDP ( $F = 6.72$ ,  $p = 0.006$ ), and unemployment ( $F = 7.40$ ,  $p = 0.004$ ). These results are statistically significant at the 5% and 1% levels, indicating a robust predictive power of renewable energy consumption. This suggests that increased reliance on renewable energy not only contributes to reducing carbon emissions but also plays a significant role in shaping economic growth and employment dynamics. The strong causal influence of renewable energy highlights its importance in driving both environmental sustainability and broader macroeconomic outcomes.

On the other hand, unemployment does not exhibit any significant causal relationship with the other variables in the model. The results for UN causing CO<sub>2</sub> ( $F = 1.54$ ,  $p = 0.241$ ), HCE ( $F = 0.45$ ,  $p = 0.645$ ), and RENE ( $F = 0.94$ ,  $p = 0.407$ ) are all statistically insignificant. This indicates that changes in unemployment levels do not help predict variations in emissions, consumption, or renewable energy usage within the given time frame.

# VISUAL ANALYSIS USING DASHBOARD



To complement the econometric analysis, a visual dashboard has been developed to present the trends and relationships among key macroeconomic and environmental variables in India over the period 2000–2025. The dashboard includes indicators such as Carbon Dioxide (CO<sub>2</sub>) emissions, Gross Domestic Product (GDP), Household Consumption Expenditure (HCE), Renewable Energy Consumption (RENE), and Unemployment Rate (UN).

This visual approach enhances the interpretability of the data by providing a clear and intuitive understanding of long-term patterns, fluctuations, and structural changes.

## GDP and CO<sub>2</sub> Emissions Trend

The dashboard clearly illustrates a consistent upward trend in both GDP and CO<sub>2</sub> emissions over the study period. As GDP increases steadily from 2000 to 2025, CO<sub>2</sub> emissions also rise correspondingly. This visual evidence strongly supports the regression results, which indicate a positive and statistically significant relationship between economic growth and environmental degradation. This trend highlights the carbon-intensive nature of economic development in India, where increased production and industrial activity contribute to higher emissions.

## Renewable Energy Consumption Trend

Renewable energy consumption shows a distinct pattern over time. Initially, there is a declining trend from 2000 to around 2015, followed by a noticeable increase in recent years. This shift

reflects policy interventions and increased investment in clean energy sources. The upward movement in renewable energy after 2015 aligns with India's sustainability initiatives and supports the regression findings that renewable energy has a significant negative impact on CO<sub>2</sub> emissions.

#### Household Consumption Expenditure (HCE)

The HCE trend appears relatively stable throughout the period, with only minor fluctuations. Unlike GDP and CO<sub>2</sub>, there is no strong upward or downward trend. This stability visually reinforces the regression result that HCE does not have a statistically significant impact on CO<sub>2</sub> emissions, indicating that consumption patterns alone do not directly drive environmental changes in the model.

#### Unemployment Trend

The unemployment rate remains relatively stable for most of the study period, with slight fluctuations. However, a noticeable decline is observed in recent years, particularly after 2020. This pattern reflects economic recovery and changing labor market conditions. The inverse relationship between unemployment and CO<sub>2</sub> emissions, as seen in the regression analysis, is also supported visually—lower unemployment corresponds with higher economic activity and emissions.

#### Impact of COVID-19 (2020 Shock)

The dashboard captures a clear structural break around the year 2020. During this period:

- GDP shows a temporary slowdown
- CO<sub>2</sub> emissions decline
- Unemployment spikes

This reflects the economic disruption caused by the COVID-19 pandemic. The subsequent recovery phase shows a rebound in GDP and emissions, along with a decline in unemployment.

## CONCLUSION

This study analyzes the relationship between household consumption expenditure and CO<sub>2</sub> emissions in India over the period 2000–2025 using econometric techniques. The findings reveal that economic growth, measured by GDP, is the most significant contributor to rising carbon emissions. This indicates that India's development path continues to rely heavily on carbon-intensive activities such as industrial production and fossil fuel-based energy use.

On the other hand, household consumption expenditure does not show a statistically significant direct impact on CO<sub>2</sub> emissions. This suggests that consumption alone is not a primary driver of environmental degradation within the model. However, its indirect role cannot be ignored, as changes in consumption patterns may still influence energy demand and sustainability outcomes over time.

Renewable energy consumption emerges as a key factor in reducing emissions, showing a strong and significant negative relationship with CO<sub>2</sub> levels. This highlights the importance of transitioning towards cleaner energy sources. Additionally, unemployment shows a negative relationship with emissions, reflecting reduced economic activity and lower energy usage during such periods.

Overall, the study concludes that carbon emissions in India are driven more by production processes and energy choices rather than consumption alone. Therefore, policy efforts should prioritize expanding renewable energy, improving energy efficiency, and promoting sustainable production systems. At the same time, encouraging environmentally responsible consumption can support long-term sustainability. Achieving a balance between economic growth and environmental protection is essential for India's future development.